Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

Palaeomagnetism of the Sarclet Sandstone (Orcadian Basin): age perspectives

K. Storhaug and K. M. Storetvedt
Institute of Geophysics, University of Bergen, 5000 Bergen, Norway

Synopsis

Palaeomagnetic results from the Sarclet Sandstone Formation, stratigraphically positioned near the base of the Old Red Sandstone (ORS) sequence of Caithness, suggest the presence of two different axes of magnetization, with Dec. 211, Inc. 33 and Dec. 191, Inc. 6 respectively. The most steeply inclined magnetization is in correspondence with results from Scottish Lower Old Red Sandstone volcanics indicating that the age of the Sarclet Sandstone dates from the topmost Silurian–lowermost Devonian. The abundant clasts of basalts in the underlying Sarclet Conglomerate give further substance to the notion that the Lower Old Red Sandstone volcanics, which have a wide distribution in central and southern regions of Scotland, also occurred in the Caithness/Moray Firth area. It is concluded therefore that the Orcadian Basin developed concurrently with the ORS basins of the Midland Valley and the Scottish Borders. It is inferred that the second magnetization (with shallow inclination) was imposed at around Middle Devonian, but owing to a relatively early silica cementation the Sarclet Sandstone was not susceptible to acquiring the "Permian" magnetization overprint that is strongly developed in the higher stratigraphic units of the Caithness Old Red Sandstone.

Introduction

In Caithness the bulk of the Orcadian Old Red Sandstone (ORS) sequence consists of flagstone and sandstone of unquestionable Middle Devonian age (cf. for example, Crampton and Carruthers 1914; Mykura 1983, for general reviews). This succession, which has a thickness of around 4 km, is preceded by the Sarclet Group consisting of nearly 450 m of conglomerate and sandstone/mudstone (Donovan *et al.* 1974). Due to the mostly barren nature of this basal group its age has been uncertain but, based on some moderately or poorly preserved spore assemblages, a topmost Lower Devonian (Emsian) origin has recently been suggested (Collins and Donovan 1977).

The Caithness ORS sediments have been subjected to a number of palaeomagnetic studies (Waage and Storetvedt 1973; Tarling *et al.* 1976; Turner *et al.* 1978; Storetvedt and Carmichael 1979; Storetvedt and Torsvik 1983). Owing to multicomponent magnetizations, including a strong "Permian" overprint, true Devonian palaeomagnetic field directions have been very difficult to extract.

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

However, in a recent analysis of some lower Caithness beds (Storetvedt and Torsvik 1983) the Sarclet Sandstone Fm. (overlying the basal Conglomerate Fm. of the Sarclet Group) appeared to have great potential for further palaeomagnetic study. The present paper reports on the data obtained in an extended palaeomagnetic survey of the Sarclet Sandstone which is exposed in sea cliffs around Sarclet Bay. The rock collection which encompasses 16 sites is first of all characterized by being thoroughly indurated through silica cementation. The sampled sandstone lies stratigraphically *c.* 300 m below the fossiliferous horizon of the Sarclet Group referred to above, but the data obtained give further substance to the age of these basal Caithness deposits.

Apart from site 11 (see Fig. 1b), which has a stratal dip of *c.* 14 degrees in a NE direction, the sampled localities are characterized by nearly horizontal layering (dip <5°). Owing to the quartzitic nature of the rock, field drilling proved very difficult and hand sampling was carried out at four of the sites (nos. 4–7). The local declination is about 9°W which is in perfect agreement with the mean declination based on a comparison between magnetic and sun compass orientations.

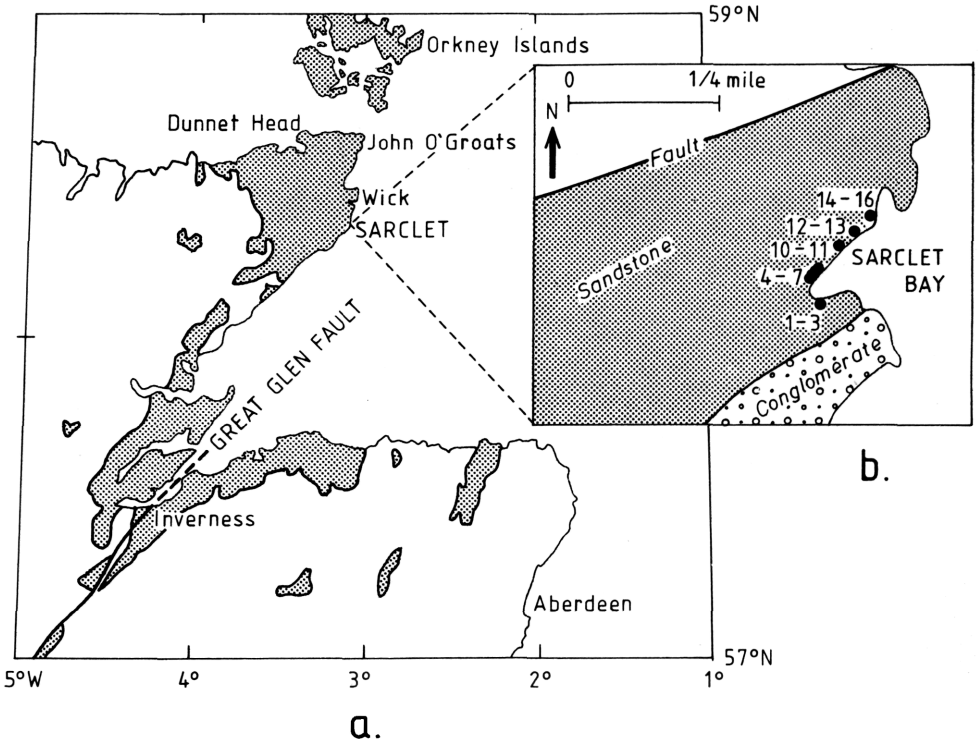


FIG. 1. Areal distribution of the Old Red Sandstone in NE Scotland (a). The investigated sites of the Sarclet Sandstone Fm. are shown in the geological sketch map (b).

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

Experimental results

80 specimens have been subjected to experimental analysis. The remanence measurements were carried out using Digico and Molspin spinner magnetometers, and a Schonsted TD-1 demagnetizer was used for the thermal treatment. Unfortunately, a stage of scattered results, coupled to a substantial rise in magnetic susceptibility, is frequently observed at temperatures >600°C. These experimental problems, which are likely to be linked to mineralogical changes, prevent the high temperature direction from being determined in many cases.

The most common category of samples is characterized by directional swings upon thermal demagnetization but without arrival at stable high temperature end points. The magnetizations erased below say 550°C are frequently displayed as straight line segments in orthogonal vector plots or, alternatively, by clustered groups of vector subtracted directions (determined from remanence vectors defining "great circle paths" on demagnetization). For a successful application of the vector subtraction technique it is important to consider demagnetization steps that make the difference vector large enough to be not unduly affected by random errors.

The least stable remanent magnetization has blocking temperatures (T_b) below *c.* 200°C. This component which is aligned along the direction of the present geomagnetic field is most likely of viscous origin and is not relevant in the present context. In the temperature interval of *c.* 200°–500°C, however, the erased vectors of many specimens define an approximately horizontal axis of magnetization striking nearly due N–S (Fig. 2). A second group of samples define stable end point magnetization as demonstrated in Figure 3. These directions may either be associated with a broad range of blocking temperatures or they become established only above 600°C.

Based on the thermal decay patterns haematite is the major remanence carrying mineral. Specimens with well-defined high temperature directions generally have "shoulder-shaped" intensity decay curves. The essential part of the fossil magnetization of such specimens is probably carried by specularite, while the magnetization of samples having more linear intensity reduction is likely to be more strongly governed by haematite pigment (cf. Collinson 1974). Chemical leaching should potentially have given a further clue to this issue, but trial experiments on a few specimens applying up to 400 hrs of hydrochloric acid (12 N) treatment, only had a minimal effect on the remanence vector. The lack of success with the chemical technique is attributed to the indurated and impermeable nature of the sediments.

Palaeomagnetic interpretation—age aspects

From the experimental data it is clear that the characteristic magnetization of the Sarclet Sandstone has a dual polarity, i.e. most specimens have superimposed

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

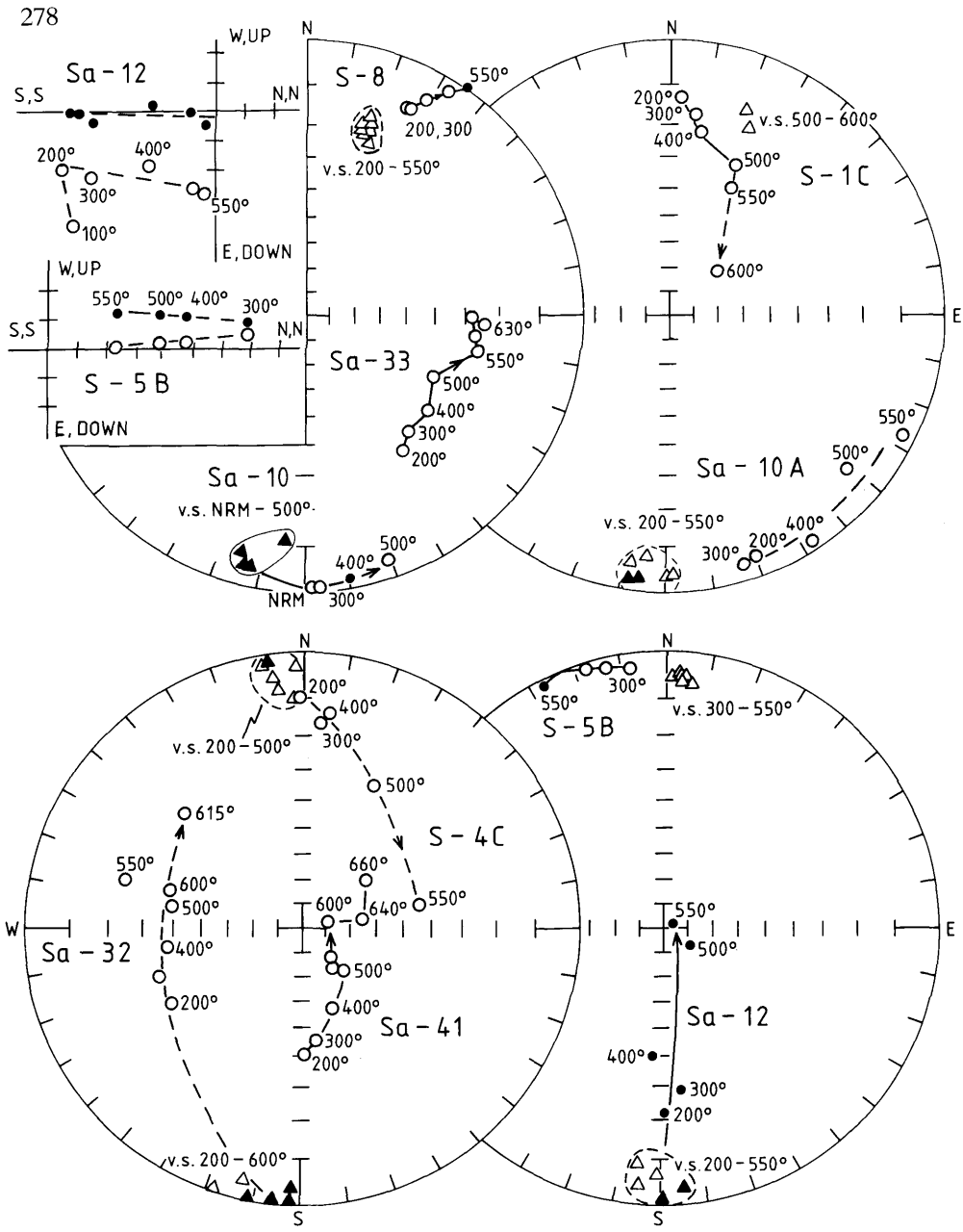


FIG. 2. Examples of specimen results from which palaeomagnetic field directions are established by vector subtraction (v.s.). Such specimens define the principal magnetization axis of the formation, Group II (see text). Remanence directions after thermal demagnetization are shown by circles; closed (open) symbols represent downward (upward) pointing magnetizations. Vector subtraction results are marked by triangles. Sa-33 and Sa-41 represent a category of samples for which there is too much component overlap to enable a successful application of vector techniques, and where stable high temperature directions were not attained (due to experimental problems).

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

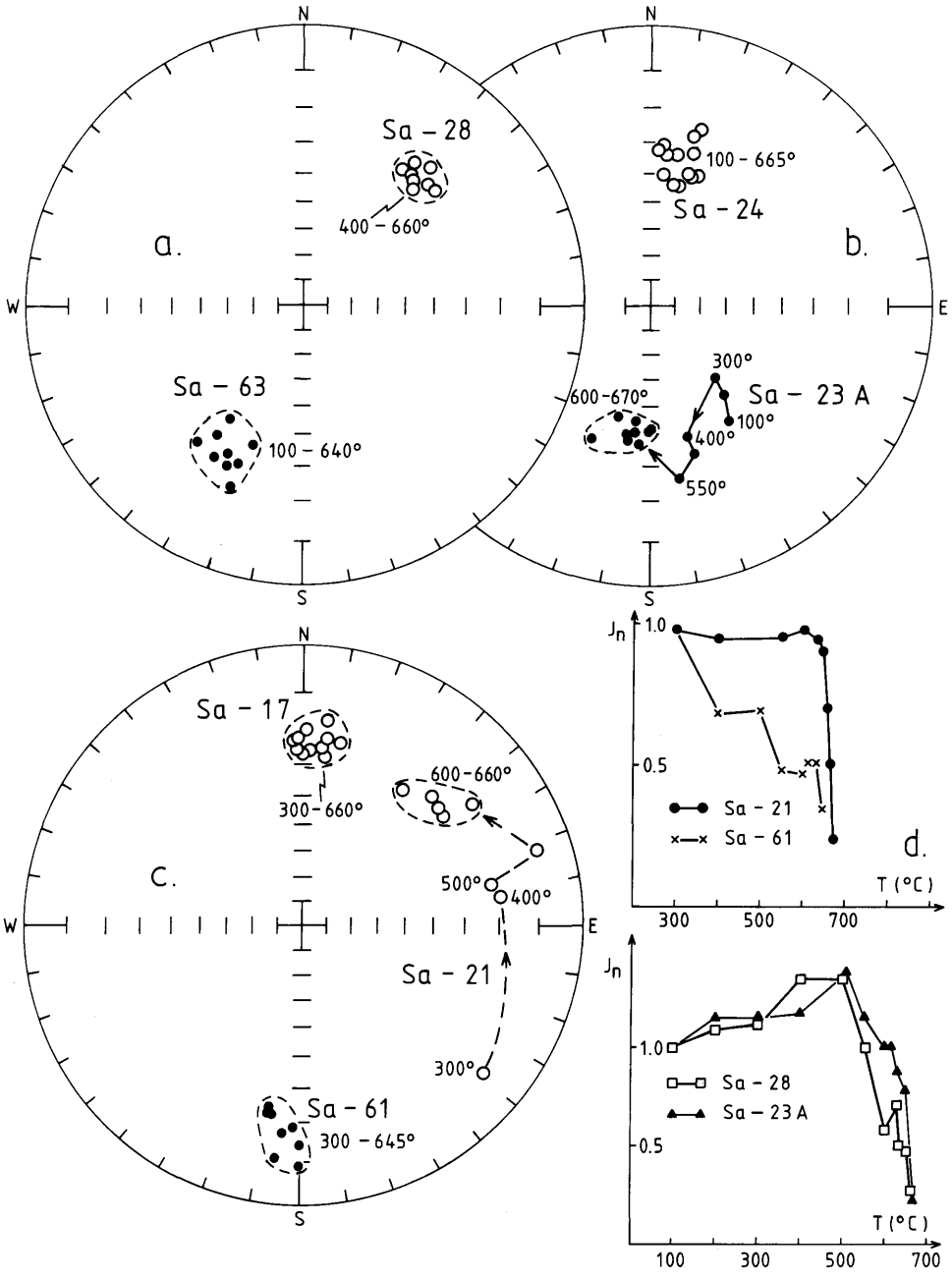


FIG. 3. Thermal demagnetization data for specimens that define stable end point magnetization. The most stable of these samples have moderately steep inclinations (cf. figs. a and b) and form a second palaeomagnetic group (Group I). See text for further details. Conventions as for Fig. 2.

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

normal and reverse magnetizations of ancient origin. The relative importance of these "anti-parallel" components varies greatly, reflecting different magnetization histories even for neighbouring specimens. However, despite the experimental problem due to overlapping component spectra, remanence directions defining consistent groupings have been determined from as much as 60 per cent of the analysed material. These data (Table 1) form the basis for the present palaeomagnetic consideration. The total directional distribution exhibits a certain "inclination" gap tentatively suggesting a subdivision into two different palaeomagnetic populations: Group I with an overall declination/inclination of 211/+33, and Group II with an average direction of 191/+6 (Fig. 4). Group II conforms closely to the results of younger (Middle-? Upper Devonian) ORS rocks in Caithness (Storetvedt *et al.* 1978; Storetvedt and Carmichael 1979; Storetvedt and Torsvik 1983). The observed declination difference between the Group II mean direction and those of the younger ORS rocks (cf. Fig. 5) is most likely linked to the certain declination smear that is observed in the individual formations. These features are interpreted as reflecting a relative clockwise rotation of Caithness due to transcurrent movement, and associated remagnetization, along the Great Glen Fault in the late Middle Devonian.

The directions of Group I match fairly well with the directions of magnetization found in Scottish Old Red Sandstone volcanics of uppermost Silurian—basal

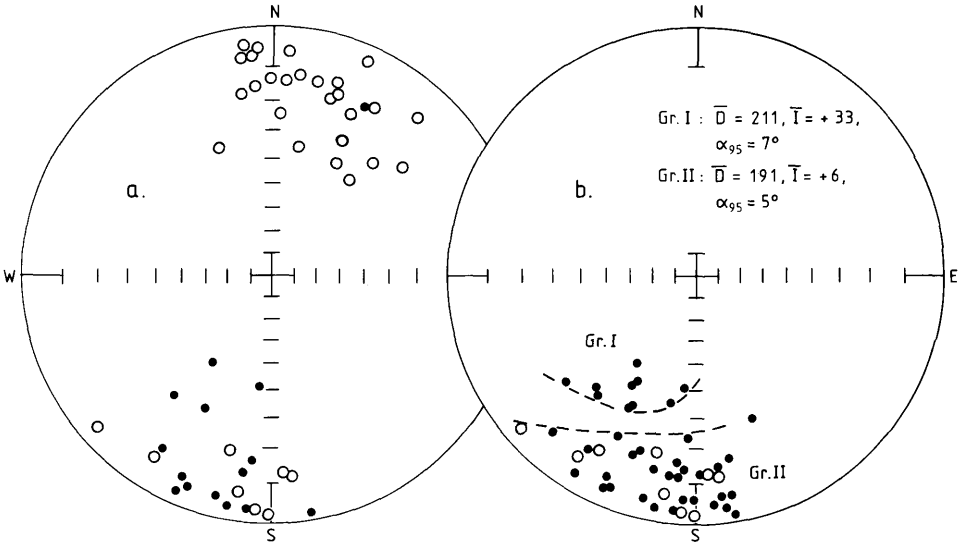


FIG. 4. Diagram a shows the total distribution of palaeomagnetic directions determined from the Sarclet Sandstone (taken from Table 1). In diagram b the normal directions have been inverted, and the combined reversed grouping shows evidence of two palaeomagnetic populations, Groups I and II. Mean declination/inclination ($\bar{D}$, $\bar{I}$) and the radius of the 95% circle of confidence (α_{95}) are given for each group.

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

TABLE 1

Palaeomagnetic results from individual specimens of the Sarclet Sandstone. Each direction, specified by declination (D) and inclination (I), represents either the mean direction over given stability interval, or the average vector subtracted direction (v.s.) for given range of demagnetization. Only site 11 results have been structurally corrected (cf. text).

Site	Specimen	D	I	Range
1	Sa-4	193	−18	v.s. 200–500
	5a	174	−12	v.s. 200–500
	5b	212	11	550–620
2	8	194	5	v.s. 200–400
	9	219	26	100–400
	10A	184	−3	v.s. 200–550
	10B	191	3	v.s. 100–500
	11B	171	2	200–500
	12	181	−2	v.s. 200–550
3	S-1A	177	−13	v.s. 200–500
	1C	19	−12	v.s. 500–600
	2C	359	−13	200–500
	3A	20	−15	v.s. 300–550
	3C	24	−4	v.s. 300–500
	4A	356	−5	200–500
	4C	354	−7	v.s. 200–550
	5B	4	−6	v.s. 300–550
4	6	353	−4	v.s. 200–500
	8	18	−17	v.s. 200–550
	10	350	−17	200–550
	11	26	−19	v.s. 200–500
	12	43	−9	v.s. 200–550
6	19	229	−4	200–550
				+ v.s. 600–660
7	21	4	−14	200–550
	23A	8	−12	200–550
	23B	13	−13	v.s. 200–600
	28B	354	−15	v.s. 200–550
8	Sa-15	351	−7	200–660
	16	30	−36	640, 650, 655, 660
	17	3	−24	300–660
	18	32	−14	200–640
	19	337	−28	500–660
9	20	27	−28	300–620
	21	51	−22	600–660
	22Aa)	202	5	v.s. 200–500
	b)	39	−38	300–645
	23A	186	41	600–665
	24	12	−35	100–665
10	27	213	−8	v.s. 100–500
	28	42	−28	400–660
11	32	188	11	v.s. 200–600
	33	184	3	v.s. 200–500
	34	29	14	300–660
12	38	204	7	v.s. 100–550
	40	204	3	v.s. 200–550
	41	189	−7	v.s. 200–550
15	57B	214	44	620–640
16	61	186	16	200–645
	63	207	28	100–640

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

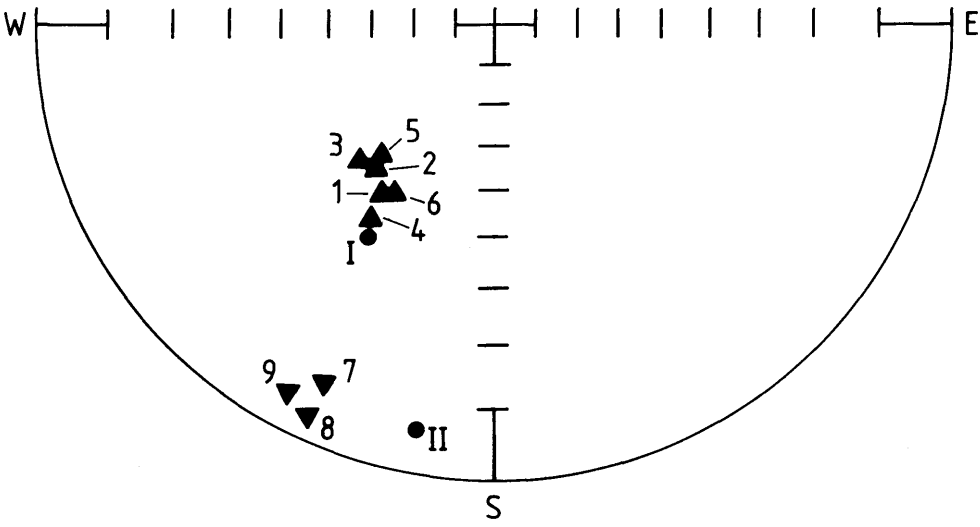


FIG. 5. The mean directions for Group I and II seen in conjunction with the average magnetization directions for other Old Red Sandstone rocks from Scotland. Reference formations 1–6 are Lower Old Red Sandstone volcanics with typical ages on the range of 390–410 m.y. These are: 1, Cheviot Hills Complex (Thorning 1974); 2, Lorne Plateau Volcanics (Latham and Briden 1975); 3, Midland Valley Lavas (Torsvik in prep.); 4, Arrochar Complex (Briden 1970); 5, Midland Valley Lavas (Sallomy and Piper 1973); 6, Garabal Hill–Glen Fyne Complex (Briden 1970). Formations 7–9 are Middle Devonian rocks as follows: 7, Duncansby Neck (Storetvedt *et al.* 1978); 8, John O'Groats Sandstone (Storetvedt and Carmichael 1979); 9, Lower Caithness Red Beds (Storetvedt and Torsvik 1983).

Devonian age (Briden 1970, Sallomy and Piper 1973, Thorning 1974, Latham and Briden 1975). The intermediate palaeomagnetic inclinations that characterize these volcanics (age around 400 m.y.) are not generally recorded in Lower ORS sediments of NW Europe. The reason for this discrepancy is most likely to be sought in: a) diverging magnetization histories and b) the apparently short lifetime of the intermediately inclined "basal" Devonian field. The volcanic rocks (mostly lavas) probably acquired their stable magnetization at around the time the rocks cooled, while the sandstone/mudstone deposits most likely underwent a more long-lasting magnetization history; iron solutions from chemical/weathering activity must have percolated through the sediments prior to final consolidation by silica cementation. We believe therefore that the most steeply inclined remanence directions displayed by the Sarclet Sandstone Fm. (Group I) represent the original magnetization acquired by these sediments while the more flat-lying magnetization (Group II) was imposed at a later stage of the Devonian basin development.

Group I constitutes samples for which a major portion of the natural remanent

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

magnetization has blocking temperatures above 600°C, suggesting the presence of a significant amount of specular haematite, while the more linear intensity reduction for Group II specimens (very few of them have magnetizations that can be followed above 600°C) can be associated with haematite pigment (Collinson 1974). Supposing that the specular haematite grains existed at the time of deposition the intermediate inclination of the ambient field at that time should have given rise to a certain inclination error. This is indeed indicated by the observations in that the overall mean direction of Group I has a slightly shallower inclination than that of the igneous reference formations (cf. Fig. 5).

Although the two mean directions are significantly different at the 95 per cent significance level a statistical discrimination between the two suggested populations, using Watson's Test (Watson 1956), is not firmly established. However, the high magnetic stability of Group I compared with that of Group II, and the fact that the two groups are reasonably separated from each other, suggest magnetizations of different origin. Using available reference data from Scotland, the Group I magnetization is regarded as the original remanence, being emplaced (primarily as a depositional magnetization) in the lowermost Devonian. Hence the early history of the Orcadian Basin is most likely to be time equivalent with other ORS basins in NW Europe. This conclusion is apparently at some variance with that of Collins and Donovan (1977), tentatively suggesting an uppermost Lower Devonian age for the Sarclet Group, though the rocks of the present study have a stratigraphic position of *c.* 300 m below the investigated fossiliferous horizon. Nevertheless, observations of macroflora in the nearby Ousdale sequence seem restricted to the Lower Devonian and Silurian (cf. Collins and Donovan 1977) and are therefore fully compatible with the palaeomagnetic considerations. According to Mykura (1983) the Sarclet Conglomerate (immediately below the Sarclet Sandstone) contains abundant clasts of basalts probably implying contemporaneous volcanism. One is tempted to conclude therefore that the Lower ORS volcanism that had an extensive distribution in southern and central regions of Scotland also affected the Caithness/Moray Firth area.

In studies of the younger Devonian strata of Caithness a considerable Permian—Mesozoic magnetization overprint has been recorded. This secondary magnetization is not present in the Sarclet Sandstone. Turner *et al.* (1976) attributed this difference to early quartz cementation in the Sarclet beds, but the present palaeomagnetic data suggest that the sediments were at least susceptible to remagnetization until *c.* Middle Devonian (Group II).

Acknowledgements

This project has been supported by the Norwegian Research Council for Science and the Humanities, Grant D.41.12.09.

Downloaded from http://sjg.lyellcollection.org/ at Monash University on June 4, 2015

References

BRIDEN, J. C. 1970. Palaeomagnetic results from the Arrochar and Garabal Hill-Glen Fyne igneous complexes, Scotland. *Geophys. J. R. Astron. Soc.* **21,** 457–70.

COLLINS, A. G. and DONOVAN, R. N. 1977. The age of two Old Red Sandstone sequences in southern Caithness. *Scott. J. Geol.* **13,** 53–7.

COLLINSON, D. W. 1974. The role of pigment and specularite in the remanent magnetism of red sandstones. *Geophys. J. R. Astron. Soc.* **38,** 253–64.

CRAMPTON, C. B. and CARRUTHERS, R. G. 1914. The geology of Caithness. *Mem. Geol. Surv. U.K.*

DONOVAN, R. N., FOSTER, R. J. and WESTOLL, T. S. 1974. A stratigraphical revision of the Old Red Sandstone of Northwest Caithness. *Trans. R. Soc. Edinb.* **69,** 167–201.

LATHAM, A. G. and BRIDEN, J. C. 1975. Palaeomagnetic field directions in Siluro-Devonian lavas of the Lorne Plateau, Scotland, and their regional significance. *Geophys. J. R. Astron. Soc.* **43,** 243–52.

MYKURA, W. 1983. Old Red Sandstone. *In* Craig, G. Y. (ed), *Geology of Scotland*, Scott. Acad. Press, Edinburgh.

SALLOMY, J. T. and PIPER, J. D. A. 1973. Palaeomagnetic studies in the British Caledonides—IV. Lower Devonian lavas of the Strathmore region, Scotland. *Geophys. J. R. Astron. Soc.* **34,** 47–68.

STORETVEDT, K. M., CARMICHAEL, C. M., HAYATSU, A. and PALMER, H. C. 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic neck, NE. Scotland: superimposed magnetizations, age of igneous activity and tectonic implications. *Phys. Earth Planet. Int.* **16,** 379–92.

STORETVEDT, K. M. and CARMICHAEL, C. M. 1979. Resolution of superimposed magnetizations in the Devonian John O'Groats Sandstone, North Scotland. *Geophys. J. R. Astron. Soc.* **58,** 769–84.

STORETVEDT, K. M. and TORSVIK, T. H. 1983. Palaeomagnetic re-examination of the basal Caithness Old Red Sandstone; aspects of local and regional tectonics. *Tectonophysics* **98,** 151–64.

TARLING, D. H., ABOU-DEEB, J. M. and EL-BATROUK, S. I. 1976. Palaeomagnetic dating of haematite genesis in Orcadian basin sediments. *Scott. J. Geol* **12,** 125–34.

THORNING, L. 1974. Palaeomagnetic results from Lower Devonian rocks of the Cheviot Hills, Northern England. *Geophys. J. R. Astron. Soc.* **36,** 487–96.

TURNER, P., VAUGHAN, D. J. and TARLING, D. H. 1978. Palaeomagnetic and mineralogical studies of Devonian lacustrine sediments from Caithness, Scotland. *Phys. Earth Planet. Int.* **16,** 73–83.

WAAGE, H. L. and STORETVEDT, K. M. 1973. The Devonian geomagnetic axis relative to the Orcadian basin, N. Scotland. *Z. Geophysik* **39,** 937–51.

WATSON, G. S. 1956. Analysis of dispersion on a sphere. *Mon. Not. R. Astron. Soc. Geophys. Suppl.* **7,** 153–59.

MS. accepted for publication 22 January 1985